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Through our work, we will seek correlations between crystallization conditions, defect contents and resolution limit of diffraction. In-situ monitoring of the evolution of defects using Atomic Force Microscopy, combined with insight into the crystallization and defect formation mechanisms, will form the basis for recommendations for changes in the crystallization parameters. Of these, purity and uniformity of the protein preparations will be quantified by liquid chromatography and light scattering. Optimization will proceed until x-ray diffraction tests yield data with resolution sufficient for biologically and medically meaningful conclusions. Accumulation of expertise during the study should significantly shorten the times required for optimization.

The overall objective of the proposed research is to develop tools, based on molecular-level monitoring of the pathways of nucleation, crystal growth and defect formation that accelerate the optimization of the crystallization conditions. These tools will be applicable to individual proteins of interest to structural biologists that do not readily yield crystals of

Task Description:	sufficient quality. This will shorten the path from microcrystals to crystals of size and perfection sufficient for the desired resolution of diffraction. These aims will be pursued through a joint and interactive application of methods typical of structural biology and crystal growth research. To a large extent, this represents a paradigm shift from the previous model of protein crystallization investigations, in which the two communities as separate entities were striving to establish a bi-directional flow of problems and ideas. Aside from demonstrations of the efficacy of a rational approach to the crystallization of important biological macromolecules and judicious selection of proteins for microgravity crystallization trials, a successful completion of the proposed program will yield the structures of the studied proteins and allow comparisons between space and earth based structures of the proteins chosen for microgravity crystallization.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Insight into the crystallization processes of biological macromolecules into crystals or aggregates can provide valuable guidelines in many fundamental and applied fields. Such insight will prompt new means to regulate protein phase transitions in-vivo, e.g., polymerization of hemoglobin S in the red cells, crystallization of crystallins in the eye lens, etc. Understanding of protein crystal nucleation will help achieve narrow crystallite size distributions, needed for sustained release of pharmaceutical protein preparations such as insulin or interferon. Traditionally, protein crystallization studies have been related to the pursuit of crystal perfection needed to improve the structure details provided by x-ray, electron or neutron diffraction methods. Crystallization trials for the purposes of structural biology carried out in space have posed an intriguing question related to the inconsistency of the effects of the microgravity growth on the quality of the crystals. List of industry research contacts: New Century Pharmaceuticals, Abbot Laboratories, Teravance Pharmaceuticals, Astra Zeneca Pharmaceuticals Who is using the results of your research? Protein crystal growers, both in industry and academia Number of times that your work has appeared in the popular media? 19 The Evening News, Azteca America Television, Houston, August 4, 2005 “UH Grad Student Develops Biosensor Chip”, KUHF/NPR Radio, Houston, June 16, 2005 5:30 p.m., and June 17, 2005, 5:30 and 7:30 a.m. “UH finds own niche in health” Houston Chronicle, Jan. 28, 2005, P. B1 The Slavi Trifovov Show, bTV, Sofia, July 7, 2004 “Using powers of physics to unlock biology's secrets” Dallas Morning News, August 04, 2003, p. E1 Time Warner 24 Hours News TV Channel, June 29, 2003 “Perfect Insulin Crystals”, Physics News Update, 636 #2, May 7, 2003 “Could UAH become the MIT of the south” The Huntsville Times, March 9, 2003, P. A1 “Columbia goes deeper for ex-UAH prof” The Huntsville Times, February 9, 2003, P. A15 “Star UAH professor departs tired of education civil war” The Huntsville Times, June 10, 2001, P. A1 “Scientists Take First Pictures of Molecules Forming Crystals Nucleus” Technology Valley, Feb. 2001, p.16 “Observing the Crystallization Process” About: Composite Materials, August 2000 “First look at a crystal's "embryo"—flat, not round”, naturalSCIENCE, News, August 3, 2000 “First Images Of Crystal Nucleation Are Unexpected” “Caught on Film: Nucleating Crystals Make Their Debut”, Microgravity News, vol. 7, No.3, Fall 2000, p.11 “Research Hopes Grow with Crystal Discovery”, The Jackson Citizen-Patriot, August 16, 2000, p. B1 “Crystal Discovery Could Lead to Sickle Cell Cure”, Seattle Times, August 12, 2000, p. A6 “UAH Breaks Crystals’ Code”, The Huntsville Times, August 9, 2000, p. A1 “Crystal Reveals Unexpected Beginnings”, Science News, August 5, 2000, p. 84 “Catching Crystals at Birth”, Nature, vol. 406, August 3, 2000, p. 464 “Protein Crystallization at NASA: Well Grounded”, Science August 6, 1999; 285: 835 Number of times that your work has appeared on a magazine cover? 2 Science vol. 299 Number 5609 February 14, 2003 ; J. Mol. Biol. Vol. 336 Number 1, Feb. 6, 2004
	Main Results Molecular processes of growth and defect formation of protein crystals Molecular-level thermodynamic and kinetic parameters for the self-assembly of apoferritin molecules into crystals The self-assembly of apoferritin molecules into crystals is a suitable model for protein crystallization and aggregation; these processes underlie several biological and biomedical phenomena, as well as for protein and virus self-assembly. We use the atomic force microscope in-situ, during the crystallization of apoferritin to visualize and quantify at the molecular-level the processes responsible for crystal growth. To evaluate the governing thermodynamic parameters, we

image the configuration of the incorporation sites, “kinks”, on the surface of a growing crystal. We show that the kinks are due to thermal fluctuations of the molecules at the crystal-solution interface. This allows evaluation of the free energy of the intermolecular bond f = 3.0 kBT = 7.3 kJ/mol. The crystallization free energy, extracted from the protein solubility, is – 42 kJ/mol. Published determinations of the second virial coefficient and the protein solubility between 0 and 40oC revealed that the enthalpy of crystallization is close to zero. Analyses based on these three values suggest that the main component in the crystallization driving force is the entropy gain of the waters bound to the protein molecules in solution and released upon crystallization. Furthermore, monitoring the incorporation of individual molecules in to the kinks, we determine the characteristic frequency of attachment of individual molecules at one set of conditions. This allows a correlation between the mesoscopic kinetic coefficient for growth and the molecular-level thermodynamic and kinetic parameters determined here. We found that step growth velocity, scaled by the molecular size, equals the product of the kink density and attachment frequency, i.e., the latter pair are the molecular-level parameters for self-assembly of the molecules into crystals.

Diffusion-limited kinetics of the solution-solid phase transition of molecular substances

For critical tests of whether diffusion-limited kinetics is an option for the solution-solid phase transition of molecular substances, or they are exclusively determined by a transition-state, we performed crystallization experiments with ferritin and apoferritin, a unique pair of proteins with identical shells but different molecular masses. We find that the kinetic coefficient for crystallization is identical (accuracy £ 7%) for the pair, indicating diffusion-limited kinetics of crystallization. Data on the kinetics of this phase transition in systems ranging from small-molecule ionic to protein and viri suggest that the kinetics of solution phase transitions for broad classes of small-molecule and protein materials are diffusion-limited.

Capillarity effects on crystallization kinetics: insulin

During layerwise growth of crystals, capillarity governs the generation of new crystal layers. Theory predicts that the line tension of the layer edge determines, via the characteristic two-dimensional capillary length Lc, the rates of generation and initial growth of the new layers. To test the correlation between Lc and the rate of layer generation, we used in-situ Tapping Mode Atomic Force Microscopy (TM-AFM) to study the generation and spreading of layers during crystallization of rhombohedral, R3, porcine insulin. We show that crystallization of this insulin form is uniquely suitable for such an investigation due to the linear kinetics of step growth it exhibits. This linear kinetics reflects the abundance of the incorporation sites along the rough steps, the lack of long-range step-step interactions, and the transport control of the growth kinetics. The kinetic coefficients are 7´10-3 and 4´10-2 cm s-1, respectively, in the absence and presence of the co-solvent acetone—somewhat high for proteins and comparable to values for inorganic systems. We show that (i) the relevant capillary length, the size of a critical quadrangular 2D nucleus Lc, is the main scaling factor for the density of growth steps, while (ii) all steps longer than Lc grow with a rate determined only by the supersaturation and independent on their length. We explain the divergence of (ii) from theoretical predictions with the high supersaturations typical of the growth of this protein system.

Evidence for the surface diffusion mechanism of solution crystallization from molecular-level observations with ferritin

We employ atomic force microscopy to monitor in situ, in real time, the molecular processes of crystallization of ferritin, a protein which has an inorganic single-crystalline core that can be varied. We determine the statistics of molecular attachment and detachment at the growth sites and find that the ratio of the fluxes in and out of the kinks is significantly lower than expected assuming direct incorporation of the molecules from the solution. Determinations of the energy barrier for incorporation yield ~ 30 kJ mol-1, significantly higher than expected for this mechanism. We conclude that attachment of molecules occurs via the surface adsorption layer. The surface coverage resulting from this mechanism is ~ 0.9, suggesting a growth mode different from the classical surface diffusion mechanism.

Molecular mechanisms of microheterogeneity-induced defect formation in ferritin crystallization

We apply in-situ atomic force microscopy to the crystallization of ferritins from solutions containing ~5% (w/w) of their inherent molecular dimers. Molecular resolution imaging shows that the dimers consist of two bound monomers. The constituent monomers are likely partially denatured resulting in increased hydrophobicity of the dimer surface. Correspondingly, the dimers strongly adsorb on the crystal surface. The adsorbed dimers hinder step growth and upon incorporation by the crystal initiate stacks of up to 10 triple and single vacancies in the subsequent crystal layers. The molecules around the vacancies are shifted by ~ 0.1 molecular dimensions from their crystallographic positions. The shifts strain the lattice and, as a consequence, at crystal sizes > 200 mm, the accumulated strain is resolved by a plastic deformation whereupon the crystal breaks into mosaic blocks 20 to 50 mm in size. The critical size for the onset of mosaicity is the similar for ferritin and apoferritin and close to the value for a third protein, lysozyme; it also agrees with theoretical predictions. Trapped microcrystals in ferritin and apoferritin induce strain with a characteristic lengthscale equal to that of a single point defect, and, as a consequence, trapping does not contribute to the mosaicity. The sequence of undesired phenomena that include heterogeneity generation, adsorption, incorporation and arising lattice strain and mosaicity in this and other proteins systems could be avoided by improved methods to separate similar proteins species (microheterogeneity), or by increasing the biochemical stability of the macromolecules against oligomerization.

Defects, strain, mosaicity, and diffraction resolution.

The growth of protein crystals, as well as any other crystal, occurs by the ordered addition of molecules. For a perfect crystal, a huge number of such additions (of the order of 1015 and higher) must occur in a strictly identical fashion. This large number awards many opportunities for misaligned attachment of single molecules, molecular aggregates (amorphous and crystalline), or other species present in the nutrient medium, as well as for short- and long-term variability of the growth process. As a result, defects ranging in scale from the molecular (mutated and conformationally-different molecules, misaligned molecules and single vacancies) through trapped impurities, clusters and oligomers, dislocations, and twinning planes, to the macroscopic (striations, occlusions, twins, blocks and grains and zones) are formed.

How dangerous are the defects? One may argue that even if the crystal contains 1 % of misplaced molecules, this will only result in 1 % decrease in diffraction intensity, or 1 % increase in the background noise. Unfortunately, this is not so. Even for the smallest, molecular level, “point” defects, it has been shown that 1) they replicate in subsequent layers during growth; 2) they cause strain with the strain field extending to 5 to 10 molecular diameters); and 3) the accumulation of strain leads to mosaicity and block structure. Are all defects dangerous? Arguments have been put forth that diffraction resolution is only affected by short-scale molecular disorder and not by mosaicity, striae, zoning, and

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block structures. There are examples in which heavily mosaic crystals diffract to high resolution. For insight, we note that the diffraction resolution is determined by the signal-to-noise ratio of high-index reflections. Since high-index crystal planes have low molecular density, larger areas of rotationally and translationally aligned molecules are needed to enhance the intensity of the reflections from these planes. Hence, crystal imperfections on the scale of microns, e.g., striae, and even tens and hundreds of microns, e.g., block structures, twins, etc., should affect the diffraction resolution obtainable from a crystal. Mosaicity, striae and block structures often lead to broader or split diffraction spots, and, hence, lower accuracy of the structure determination [58,59]. However, if the crystal consists of a few large blocks, the beam in a X-ray diffraction experiments can be focused on only one of these blocks, and high resolution structure determinations can still be achieved.

The various defects that may be present in a protein crystal have been classified as:

I.…Sub-molecular level defects: mutated molecules, conformational changes;

II. Rotational and translational lattice defects:

- variability of the rotational orientation of the molecules in the crystals; - vacancies or molecules out of their lattice positions;

III. Impurity-related defects, caused by other species present in the solution;

IV. Linear and planar defects: dislocations, twins, sector, grain and block boundaries;

V. Striations, occlusions;

VI. Incorporation of microcrystals.

Thermodynamics of protein crystallization

Intermolecular interactions in solution

Light scattering was applied to study apoferritin solution. Apoferritin and ferritin crystals grow well in solutions buffered at pH around or above 5, in the presence of a divalent cation Mn2+, Cd2+, Ni2+, Co2+, and others. A typical set of conditions providing for high crystal quality is 0.2 M Na acetate buffer at pH 5 with 2.5 % (w/v) CsSO4, leading 0.12 M Cd2+. Since the pI of these proteins is about 4.5, at pH 5 they are negatively charged and the intermolecular interactions are strongly affected by the cations in the solution. When plotted against the electrolyte concentration, the dimensionless second osmotic virial coefficient B2 in apoferritin solutions that only contain the Na+ ions exhibits a minimum at [Na+] between 0.1 and 0.15 M. The value of B2 at the minimum of ~ 4 is equal to that expected for non-interacting hard spheres, indicating that at these [Na+], electrostatic repulsion is completely suppressed. The ascending branch of this dependence is a manifestation of a surprisingly strong repulsion between the molecules at electrolyte concentrations about and above 0.2 M, where electrostatic interactions are insignificant. This strong deviation form the predictions of the DLVO theory was attributed to the water structuring, enhanced by the accumulation of hydrophilic counter-ions around the apoferritin molecules, giving rise to so-called hydration forces.

The interaction potential due to the hydration forces has been described as a function of the distance r from a molecule of diameter 2a as [134,135]

. Here, f0 and L are the parameters for the surface energy density and decay length respectively. Choosing decay length L = 2´0.72 nm = 1.44 nm (twice the diameter of a hydrated sodium ion), and f0 = 12.5 mJ/m2, in the middle of the range determined by surface force measurements, the values of the second virial coefficient at [Na+] > 0.15 M were reproduced.

The addition of even 0.01 M Cd2+ leads to a drop of the virial coefficient in a solution with 0.2 M Na+ from the relatively large positive value typical ofthe hydration repulsion, to about zero. Note that even these low Cd2+ concentrations are orders of magnitude higher than the apoferritin concentrations (in the micromolar range in a 1 mg/mL solution of this protein with Mw = 450,000 Da). However, further increases of the [Cd2+] up to 0.22 M = 2.5 % (w/v) (the value typically used in crystallization trials) did not lower the value of B2 further and it remained around zero. We interpret the action of Cd2+on the molecular interactions in the following way. The strong coordination bond that Cd2+ establishes between two apoferritin molecules is reflected in the potential as a deep minimum limited to distances of about 0.2 nm, i.e., comparable to the size of the Cd2+ ion. If, in addition to this effect, Cd2+ would also destroy the hydration shell around each molecule, this deep minimum would lead to highly negative B2 values. The closeness of the actual values to zero indicates that the repulsive hydration shells are present even with the Cd2+ in the solution. This results in a potential around an apoferritin molecule in a crystallizing solution containing 0.2 M Na+ and 0.22 M Cd2+.

Thus, the combination of Na+ and Cd2+ in crystallizing solutions of apoferritin leads to the non-monotonic potential of intermolecular interactions. This shape of the potential significantly enhances the selectivity of the crystallization process by allowing only a few of the attempted collisions between the molecules, or between single molecules and existing clusters and crystallites, to proceed. This selectivity may be the factor underlying the robustness of the crystallization conditions—rather significant changes in the concentrations of the two ions, and not lead to decaying crystal quality.

Solvent entropy is a main contributor to the free energy of protein crystallization

We show with six proteins that trapping and release of the water molecules upon crystallization is a determinant of the crystallization thermodynamics. With HbC, a strong retrograde solubility dependence on temperature yields a high positive enthalpy of 155 kJ mol-1, i.e., crystallization is only possible because of the huge entropy gain of 610 J mol-1K-1, stemming from the release of up to 10 water molecules per protein intermolecular contact. With apoferritin, the enthalpy of crystallization is close to zero. The main component in the crystallization driving force is the entropy gain due to the release upon crystallization of two water molecules bound to one protein molecules in solution. With both proteins, the density of the growth sites imaged by AFM is in excellent agreement with a calculation using the crystallization free energy. With lysozyme, the entropy effect due to the restructuring of the water molecules is negative. This leads to higher solubility.

Experimental determination of phase diagrams

For insight into the thermodynamics and phase behavior in concentrated protein solutions, we study the liquid-liquid

phase separation with lysozyme. We determine independently the binodal and spinodal lines, and the second virial coefficient of the protein at 275 < T < 295 K. From these data, we determine the protein chemical potential and osmotic pressure for concentrations as high as 320 mg mL-1 in the above temperature range. We find that for this protein the enthalpy of the liquid-liquid separation vanishes at the critical temperature Tc, and is comparable to and may exceed the crystallization enthalpy (~ 65 kJ mol-1) at lower T's. The enthalpy of the pair interactions averaged over all polar angles is significantly lower; this comparison suggests structuring of the dense liquid. We propose that the pair of parameters (molecular volume, second virial coefficient) may be an adequate predictor of the phase behavior of solutions of proteins with relatively simple interaction potentials.

Monte-Carlo Simulations of phase diagrams

We apply Monte Carlo simulation techniques to the investigation of phase diagrams associated with large biological molecules in solution. The system is modeled using short-range two-body intermolecular potentials exhibiting multiple extrema. We show that the introduction of a local repulsive maximum or a secondary attractive minimum in these potentials has dramatic effects on the phase diagram. Both stable and metastable liquid-liquid separation curves are driven to lower temperatures, the sensitivity of the solubility curve (liquidus) to temperature is significantly reduced, the enthalpy of crystallization is significantly diminished, metastable liquid-liquid separation may become stable and vice versa, and both low and high density solid phases may be observed.

Nucleation of protein crystals and dense liquid droplets

The nucleation mechanism of the solution—crystals phase transition. We investigated the nucleation of crystals of the protein lysozyme from aqueous solutions using a novel technique that allows direct determinations of homogeneous nucleation rates. The method is applicable to studies of crystallization, aggregation, and similar first-order phase transitions in solutions of proteins or other soluble slow-growing materials with temperature-dependent solubility. To obtain reproducible statistical characteristics of the intrinsically random nucleation process, a large number of simultaneous trials take place under identical conditions.

The nucleation theorem allows determination of the changes of the sizes of the critical cluster with increasing supersaturation as (10 or 11) ® (4 or 5) ® (1 or 2). Furthermore, we observe that the existence of a second liquid phase at high protein concentrations, Fig. 9, strongly affects crystal nucleation kinetics: (i) Crystal nucleation rates are lower than expected in the phase region of liquid-liquid demixing. (ii) In the immediate proximity of this region, nucleation rates vary by factors of up to two in identical experiments. Since experiments discussed in next subsection show a sharp rate maximum in this region, we attribute this kinetic instability to minor shifts of the experimental conditions toward or away from the phase boundary.

Dense liquid precursor for the nucleation of ordered solid phases from solution.

We showed that the presence of a liquid-liquid (L-L) phase boundary hidden below the solubility line in the phase diagram of the protein solution, has a profound effect on the rate of homogeneous nucleation J. In the vicinity of the L-L boundary (the respective TL-L are marked with vertical lines) J passes through a maximum, similar to predictions of some of the simulations and theory. These and other recent results on the kinetics of homogenous nucleation of protein crystals indicate that under a very broad range of conditions the nucleation of lysozyme crystals occurs as a superposition of fluctuations along the order parameters density and structure. Depending on whether the system is above or below its liquid-liquid coexistence line, a density fluctuation may never or may selectively lead to the formation of a dense liquid droplet; in the former case the high density region, the "quasi-droplet", is metastable also with respect to the dilute solution. In both cases, the molecules contained in the high-density region may attain an ordered arrangement, i.e., a structure fluctuation is superimposed on the density fluctuation and a crystalline nucleus obtains. This outlook on the nucleation of ordered solids from dilute phases suggests that the rate of nucleation can be controlled either by shifting the phase region of the dense liquid phase, or by facilitating the structure fluctuations within a dense liquid droplet or quasi-droplet. Results from literature indicate that the proposed two-step nucleation mechanism and the related tools for nucleation control may be applicable to the formation of crystalline and non-crystalline ordered solid phases of other, protein and non-protein materials, from solution.

Control of the nucleation rate by controlling the concentration fluctuations.

Comparing data in the same supersaturation ranges recorded away from the L-L boundary, we find that the rate of nucleation is enhanced by factors of 6 to 20 . The found correlation allows control of the nucleation rate of protein crystals by using additives that enhance or suppress the concentration fluctuation in protein solutions supersaturated with respect to an ordered solid phase. Glycerol and polyethylene glycol (PEG) (these reagents that shift this phase boundary, and do not specifically bind to proteins), significantly suppress (glycerol) or enhance (PEG) the crystal nucleation rates at T ³ TL-L(no additive), i.e., in the region of coexistence of the dilute solution and solid phases, where control of nucleation of the solid phase is sought. Determinations of the phase diagrams in the presence of these additives and of their effects of the protein-protein interrelations in solution indicate that glycerol enhances the repulsion between the molecules, while PEG brings about stronger attraction. Correspondingly, glycerol suppresses the concentration fluctuations, while PEG enhances them. The effects of the two additives on the concentration fluctuations correlate with their effects on the nucleation rate. This correlation supports the suggested two-step mechanism of nucleation of protein crystals, and provides for a direct control mechanism of the nucleation rate. The control mechanism does not require changes in the protein concentration, the acidity and ionicity of the solution, temperature or protein concentration. The effects of the two additives on the phase diagram depend on their concentration and this provides for further tuning of nucleation rates.

Smooth transition from nucleation to spinodal decomposition in phase separating protein solutions

For insight into the structure and dynamics of phases emerging upon crossing the metastability/instability boundary we monitor with optical microscopy, in real time and in real space, the generation of a dense liquid phase in high-concentration solutions of the protein lysozyme after temperature quenches into thermodynamically defined metastable and unstable regions. We show with this system, which is a poor fit to mean-field assumptions, that the evolution of the structure factor during nucleation is similar to that during spinodal decomposition and reveals no singularity predicted upon crossing the metastability boundary. We introduce two new kinetic definitions of the metastability/instability boundary that yield values within ~ 1.5 K, i.e., the boundary appears as an area rather than a line, which is near and above the thermodynamic prediction. Delay times for the appearance of the new phase in the unstable regime are significant, i.e., new-phase growth is hindered by kinetic barriers. While our results agree with

	predictions of the non-mean-field theories of phase transformations, the experimentally observed behavior is richer than the one envisioned by theory.
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